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MATHEMATICAL MYSTERIES

We Know Only Three Wilson Primes

Every prime satisfies Wilson’s theorem. Only three known primes go one step further — and nobody knows if a fourth exists.



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Prime numbers are already unusual. They become less frequent as numbers grow, resist simple patterns, and continue to generate questions that look elementary but remain unresolved.

Yet hidden among the primes is a far rarer species, the **Wilson primes**.

Only three are known:
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$$5,\ 13,\ 563$$

What makes them special can be stated surprisingly simply.

For every prime p, **Wilson's theorem** guarantees that $(p-1)! + 1$ is divisible by p:

$$p \mid ((p-1)! + 1)$$

But **Wilson primes** satisfy a rarer condition: the same number is divisible by p^2:

$$p^2 \mid ((p-1)! + 1)$$

That extra factor of p is **extraordinarily rare**.

Only 5, 13, and 563 are known to satisfy this condition. No fourth example has been found, despite computational searches extending all the way to
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$$2 \times 10^{13}$$

That is **20 trillion**.

So the mystery is not whether Wilson's theorem works. It works for every prime. The mystery is why, in a tiny handful of cases, it works one power of p better than it has to.

To understand why that is so remarkable, we first need to see what **Wilson's theorem** says — and why it works.

A Theorem Every Prime Obeys

Take a positive integer n. Its factorial is
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$$n! = 1 \times 2 \times 3 \times \cdots \times n$$

For example,
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$$4! = 1 \times 2 \times 3 \times 4 = 24$$

Now suppose p is prime and consider
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$$(p-1)! + 1$$

Something remarkable always happens: this number is divisible by p.

In modular notation,
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$$\boxed{(p-1)! \equiv -1 \pmod{p}}$$

This is **Wilson's theorem**.

For $p = 5$,
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$$4! + 1 = 25$$

which is divisible by 5.

For $p = 7$,
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$$6! + 1 = 721 = 7 \times 103$$

For $p = 11$,

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$$10! + 1 = 3\,628\,801$$

and
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$$3\,628\,801 = 11 \times 329\,891$$

Again, the prime divides the factorial plus one.

But **Wilson's theorem** is a **two-way statement**. For every integer $n > 1$,
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$$\boxed{n \text{ is prime} \iff (n-1)! \equiv -1 \pmod{n}}$$

The implication works in **both directions**. If n is prime, the congruence holds. And if the congruence holds, then n must be prime.

So this is not merely a property of primes. It is a **characterization of primality**.

To see the characterization in action for a composite number, take $n = 6$:
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$$5! + 1 = 121 = 11^2$$

Although **121** is a perfect square, it is not divisible by **6**. Therefore,
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$$5! \not\equiv -1 \pmod 6$$

The theorem works perfectly. **But why?**

Why Wilson's Theorem Works

The explanation is one of those number-theory arguments that is much simpler than the statement initially suggests.

Consider the nonzero residues **modulo 7**:
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$$1,\ 2,\ 3,\ 4,\ 5,\ 6$$

Each has a multiplicative inverse modulo 7. This means that for each number, there is another number among 1, 2, ..., 6 such that multiplying the two leaves a remainder of 1 when divided by 7.

For example,
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$$2 \cdot 4 = 8 \equiv 1 \pmod 7$$

while
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$$3 \cdot 5 = 15 \equiv 1 \pmod 7$$

There are two exceptions to the ordinary pairing. The number 1 is its own inverse, and so is 6:
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$$1 \cdot 1 \equiv 1 \pmod 7$$

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$$6 \cdot 6 = 36 \equiv 1 \pmod 7$$

Since
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$$6 \equiv -1 \pmod 7$$

these two exceptional numbers are precisely 1 and −1.

Now write
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$$6! = 1 \cdot 2 \cdot 3 \cdot 4 \cdot 5 \cdot 6$$

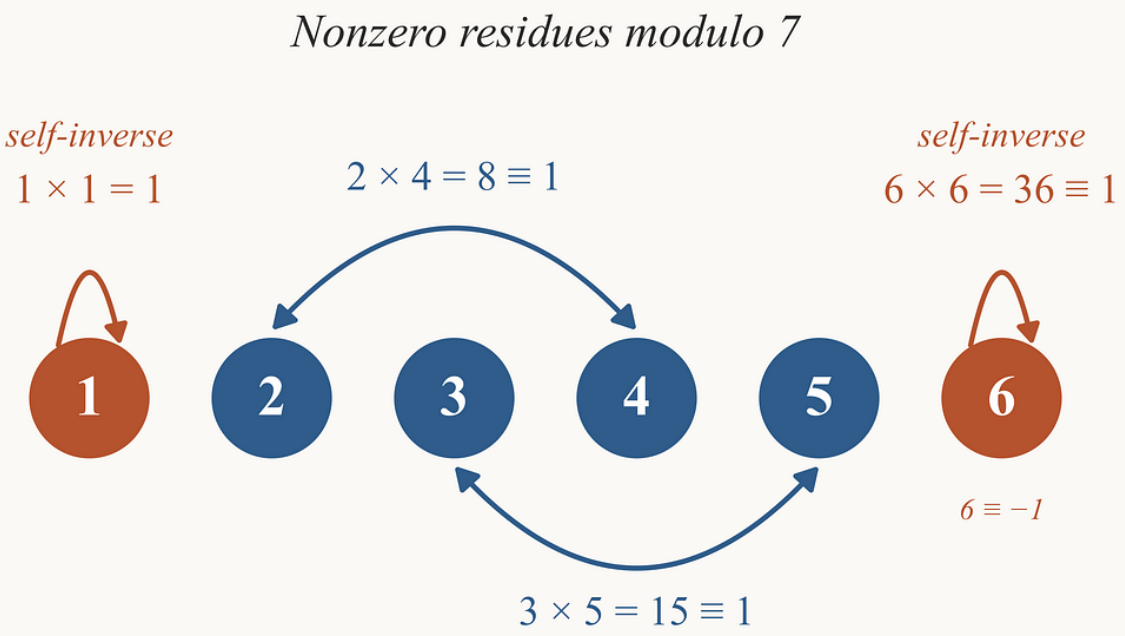
We can group the inverse pairs:
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$$6! = 1 \cdot (2 \cdot 4) \cdot (3 \cdot 5) \cdot 6$$

Modulo 7,
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$$6! \equiv 1 \cdot 1 \cdot 1 \cdot 6 \equiv 6 \equiv -1 \pmod 7$$

That is **Wilson's theorem in action**.
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$$6! \equiv 1 \times (2\times4) \times (3\times5) \times 6 \equiv 1 \times 1 \times 1 \times 6 \equiv -1 \pmod 7$$

The inverse-pair structure behind Wilson's theorem, illustrated modulo 7: 2 pairs with 4, 3 pairs with 5, while 1 and 6 are self-inverses, leaving 6! ≡ −1 mod 7.

The same argument works for every odd prime p. The case $p = 2$ is immediate.
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$$1, 2, \ldots, p-1$$

every residue has a unique multiplicative inverse modulo p. Almost all of them form distinct pairs whose products are 1.

The only self-inverses satisfy
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$$x^2 \equiv 1 \pmod{p}$$

or
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$$(x-1)(x+1) \equiv 0 \pmod{p}$$

Because p is prime, this forces
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$$x \equiv 1$$

or
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$$x \equiv -1 \pmod{p}$$

Everything therefore pairs away except 1 and −1, leaving
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$$(p-1)! \equiv 1 \cdot (-1) \equiv -1 \pmod{p}$$

So we understand exactly why every prime supplies one factor of p in
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$$(p-1)! + 1$$

That fact will matter later.

A Theorem With A Long History

Wilson's theorem has a history extending much further back than its name suggests. The result was known in some form to the medieval mathematician **Ibn al-Haytham**.



Ibn al-Haytham (c. 965–c. 1040), mathematician, astronomer, and physicist of the Islamic Golden Age. Cropped from the frontispiece of Johannes Hevelius's Selenographia (1647), after Adolph Boÿ and engraved by Jeremias Falck. Public domain. Wikimedia Commons.

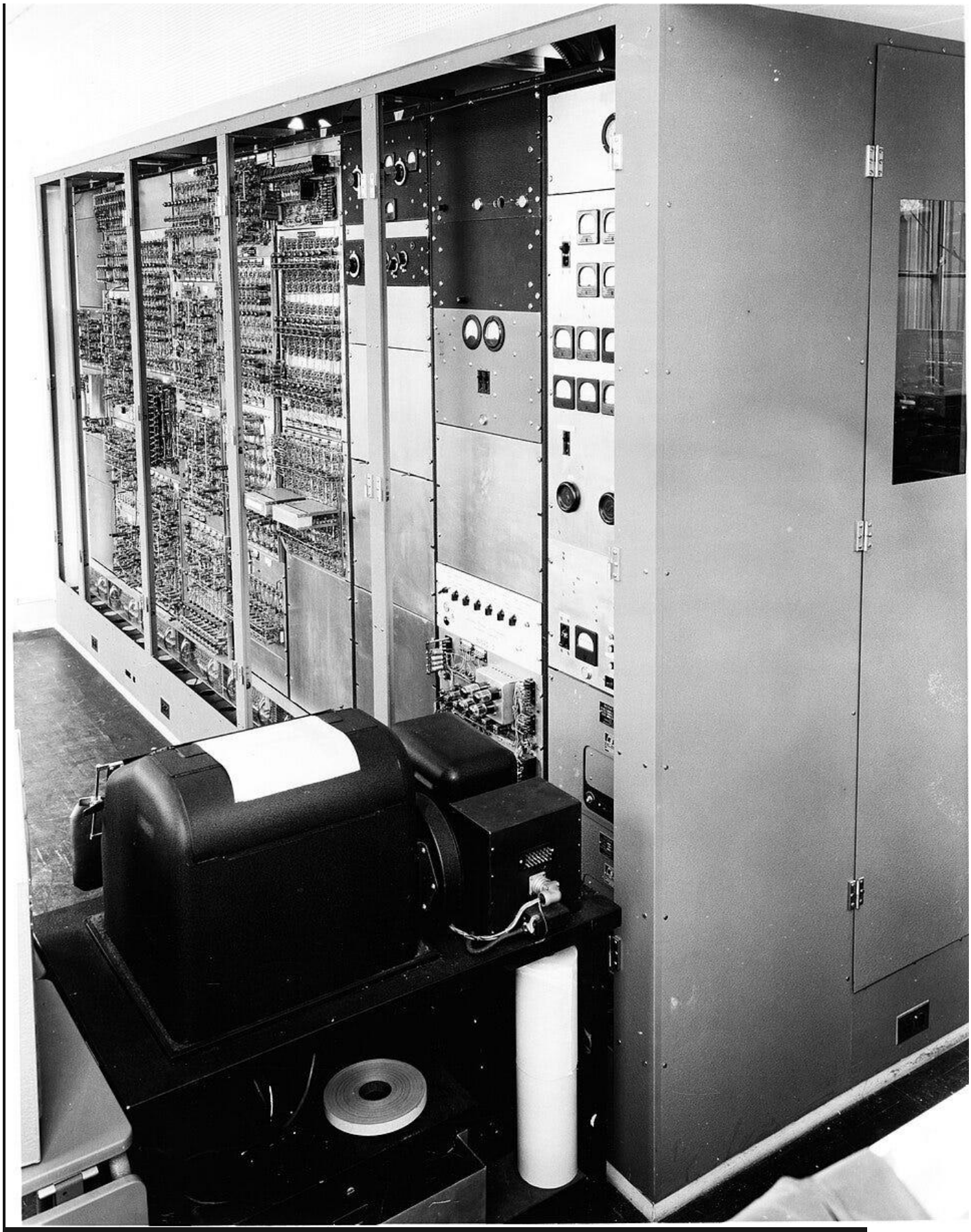
Centuries later, **Edward Waring** published it in **1770** and attributed it to his student **John Wilson**. Joseph-Louis Lagrange supplied a proof the following year.



John Wilson (1741–1793), British mathematician and lawyer, portrayed around 1780. Unknown artist. Public domain. Wikimedia Commons.

The rarer primes associated with the theorem entered the computational era much later. In **1953**, **Karl Goldberg** used the **Standards Eastern Automatic Computer (SEAC)** at the **U.S. National Bureau of Standards** to search all primes below 10,000. The machine, which had begun operating only a few years earlier, was one of the pioneering stored-program electronic computers. Goldberg's search revealed **563** as the **third known Wilson prime**.

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The Standards Eastern Automatic Computer (SEAC) at the U.S. National Bureau of Standards, photographed in 1950. The foreground shows the operator's control table with teletype equipment used for direct input and output. National Institute of Standards and Technology Digital Collections, Gaithersburg, MD 20899. Public domain. Wikimedia Commons.

By then, the apparently simple theorem had acquired a new question.

What happens if the divisibility goes one step further than Wilson's theorem requires?

When Divisibility By p Becomes Divisibility By p²

We already know the basic distinction. Wilson's theorem guarantees one factor of p:
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$$p \mid ((p-1)! + 1)$$

That means $(p - 1)! + 1$ contains at least one factor of p.

Wilson primes are the exceptional cases in which another factor of p appears:
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$$p^2 \mid ((p-1)! + 1)$$

Equivalently,
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$$\boxed{(p-1)! \equiv -1 \pmod{p^2}}$$

A prime satisfying this additional p^2 divisibility condition is called a **Wilson prime**.

Wilson's theorem guarantees one factor of p. Nothing in its proof forces this second factor. That is what makes the condition mysterious.

The First One Is Already In Front Of Us

We have actually calculated the first Wilson prime without noticing it.

For $p = 5$,
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$$4! + 1 = 25$$

But
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$$25 = 5^2$$

Therefore,
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$$5^2 \mid 4! + 1$$

So 5 is a Wilson prime.

Now compare this with 7:
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$$6! + 1 = 721 = 7 \times 103$$

Wilson's theorem gives us one factor of 7, exactly as expected.

But
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$$7^2 = 49$$

does not divide 721. Thus 7 is not a Wilson prime.

The distinction is tiny to write:
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$$5^2 \mid 4! + 1$$

whereas
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$$7^2 \nmid 6! + 1$$

But that extra factor turns out to be remarkably difficult to obtain.

Then Comes 13

The next Wilson prime is 13.

We have

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$$12! + 1 = 479\,001\,601$$

Since
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$$13^2 = 169$$

and
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$$479\,001\,601 = 169 \times 2\,834\,329$$

we obtain
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$$13^2 \mid 12! + 1$$

So the first two examples are
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$$5,\ 13$$

At this point, nothing looks particularly alarming. Perhaps such primes simply appear occasionally.

Then we keep searching.

The Third One Is 563

The next Wilson prime is
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$$563$$

There are **96 primes strictly between 13 and 563**. Every one of them fails. Then, suddenly,

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$$562! \equiv -1 \pmod{563^2}$$

So **563 works**.

And after **563**, the **numerical desert** becomes vastly larger.

A large-scale computation by **Edgar Costa**, **Robert Gerbicz**, and **David Harvey** searched for Wilson primes all the way to
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$$2 \times 10^{13}$$

without finding another one. **Twenty trillion**.

And still the known list contains only three numbers.

This is where the problem stops feeling like an odd divisibility trick and starts feeling like a genuine mathematical mystery.

The Wilson Quotient Makes The Mystery Visible

There is a useful way to reformulate the condition.

For a prime p, define the **Wilson quotient**
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$$W_p = \frac{(p-1)!+1}{p}$$

Wilson's theorem guarantees that W_p is an integer.

For $p = 5$,
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$$W_5 = \frac{4!+1}{5} = 5$$

Therefore,
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$$W_5 \equiv 0 \pmod 5$$

For $p = 7$,
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$$W_7 = \frac{6!+1}{7} = 103$$

and
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$$103 \equiv 5 \pmod 7$$

So 7 fails.

A prime is therefore a Wilson prime exactly when
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$$\boxed{W_p \equiv 0 \pmod p}$$

The first few residues are
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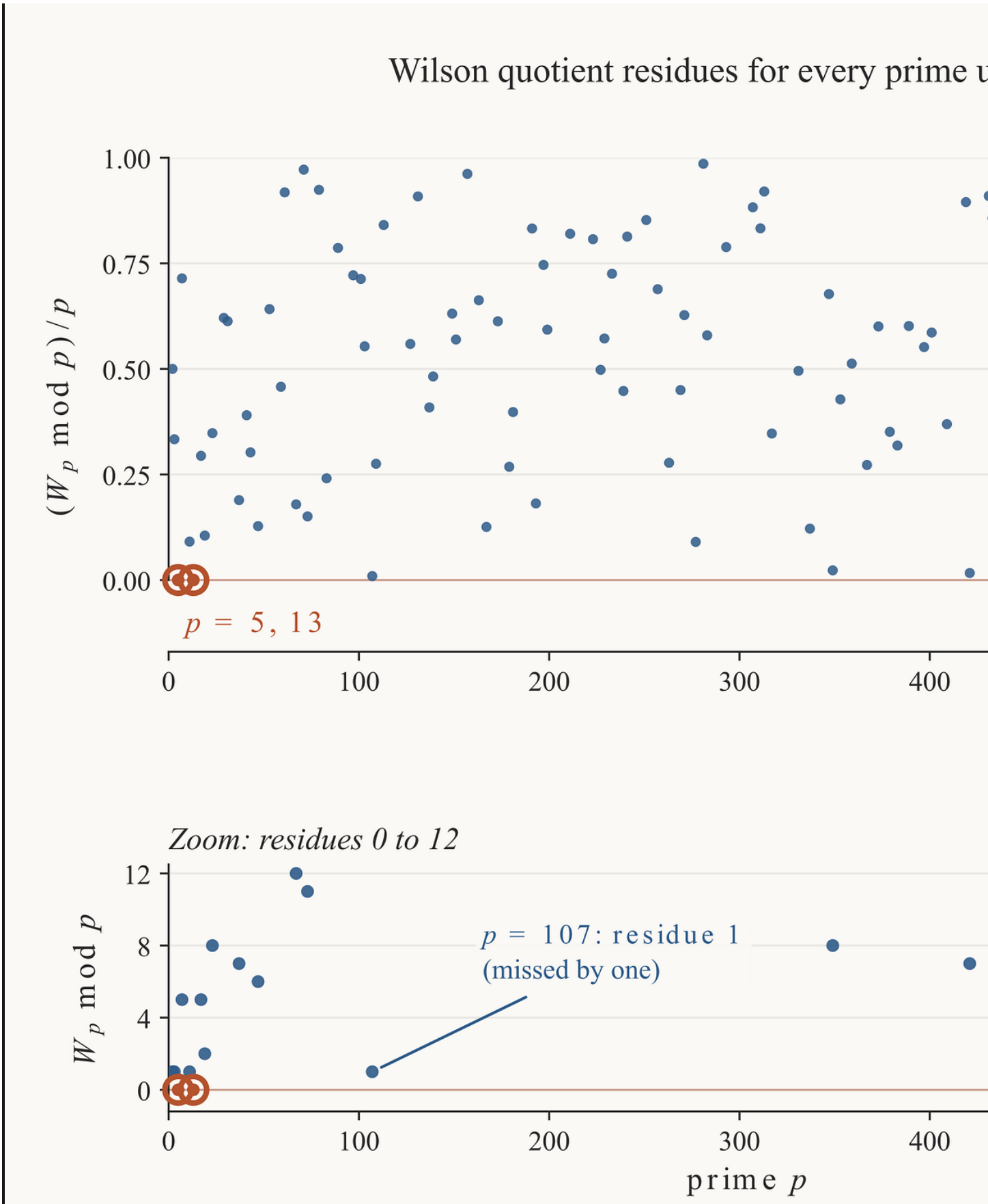
p	**5**	7	11	**13**	17	19	23	29	31	37
$W_p \bmod p$	**0**	5	1	**0**	5	2	8	18	19	7

Wilson quotient residues for the first several primes. A Wilson prime occurs exactly when $W_p \equiv 0 \bmod p$, highlighted here for $p = 5$ and $p = 13$.

Already, the zeros stand out.

But the picture becomes much more striking if we continue through all primes below 600.

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Wilson quotient residues for every prime u
1.00
0.75
0.50
0.25
0.00
$(W_p \bmod p)/p$
$p = 5, 13$
0
100
200
300
400
Zoom: residues 0 to 12
12
8
4
0
$W_p \bmod p$
$p = 107$: residue 1
(missed by one)
0
100
200
300
400
prime p

Wilson quotient residues for all primes up to 600. The three known Wilson primes, p = 5, 13, and 563, are exactly the points where W_p mod p equals zero; the lower panel magnifies residues 0–12: points that seem to touch zero above, such as p = 107, are not zeros.

The graph shows the Wilson-prime condition directly.

Among all those primes, only three points touch zero: **5, 13, 563**. Everything else misses.

Why Rare May Still Mean Infinite

There is a simple heuristic that gives some intuition for this rarity.

Once p is known to be prime, Wilson's theorem guarantees
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$$(p-1)! + 1 = pW_p$$

For p to be a Wilson prime, we need
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$$W_p \equiv 0 \pmod{p}$$

There are p possible residues modulo p:

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$$0, 1, 2, \ldots, p-1$$

If, and this is only a heuristic, the residue of W_p behaved roughly like a random choice among these possibilities, then only one residue would work.

So the probability that a given prime behaves like a Wilson prime would be roughly
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$$\frac{1}{p}$$

As p grows, this becomes extremely small.

But small probabilities accumulated over infinitely many primes can behave in surprising ways.

Under this model, the expected number of Wilson primes up to some large scale is related to
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$$\sum_{p \leq x} \frac{1}{p}$$

The sum of the reciprocals of the primes diverges, although extraordinarily slowly. More precisely,

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$$\sum_{p \leq x} \frac{1}{p} \sim \log \log x$$

The **double logarithm** grows extraordinarily slowly.

To see just how slowly, consider an almost absurd scale. Suppose $x = 10^{80}$, roughly the commonly quoted order of magnitude for **the number of atoms in the observable universe**. Under the same $1/p$ heuristic, the expected number of Wilson primes from 5 all the way up to that enormous limit would still be only about **4.6**.

So “infinitely many” does not mean “frequently encountered.” An infinite family can remain extraordinarily sparse.

The heuristic also happens to agree surprisingly well with the total number found so far. From 5 up to the verified search limit 2×10^{13}, it predicts about **2.85 Wilson primes**. We know exactly **three**: **5**, **13**, and **563**.

That agreement should not be taken too seriously — the model is only a heuristic — but it gives some confidence that the scale of the estimate is reasonable.

To estimate how many Wilson primes we might expect between two limits x and y, we subtract the corresponding double logarithms. This gives

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$$\mathbb{E} \approx \log\left(\frac{\log y}{\log x}\right)$$

This offers a curious possibility. Wilson primes could be **infinite in number** and still become **fantastically sparse**.

Is Finding Nothing Up To 20 Trillion Surprising?

We can make the heuristic concrete. Take

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$$x = 563$$

and

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$$y = 2 \times 10^{13}$$

Then
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$$\log\left(\frac{\log(2\times10^{13})}{\log 563}\right) \approx 1.58$$

So the crude model would suggest roughly **1.58 Wilson primes** in that interval. The actual number found was zero.

At first, that may seem like evidence against the model.

But suppose we go one step further and treat the expected number as the mean of a **Poisson distribution**. The probability of observing zero events would then be approximately
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$$e^{-1.58} \approx 0.21$$

About 21%.

So even under a heuristic that ultimately favors infinitely many Wilson primes, finding none between 563 and 2×10^{13} is not especially shocking.

It is unlucky.

But it is quite plausible.

And the same model illustrates just how slowly another discovery might arrive.

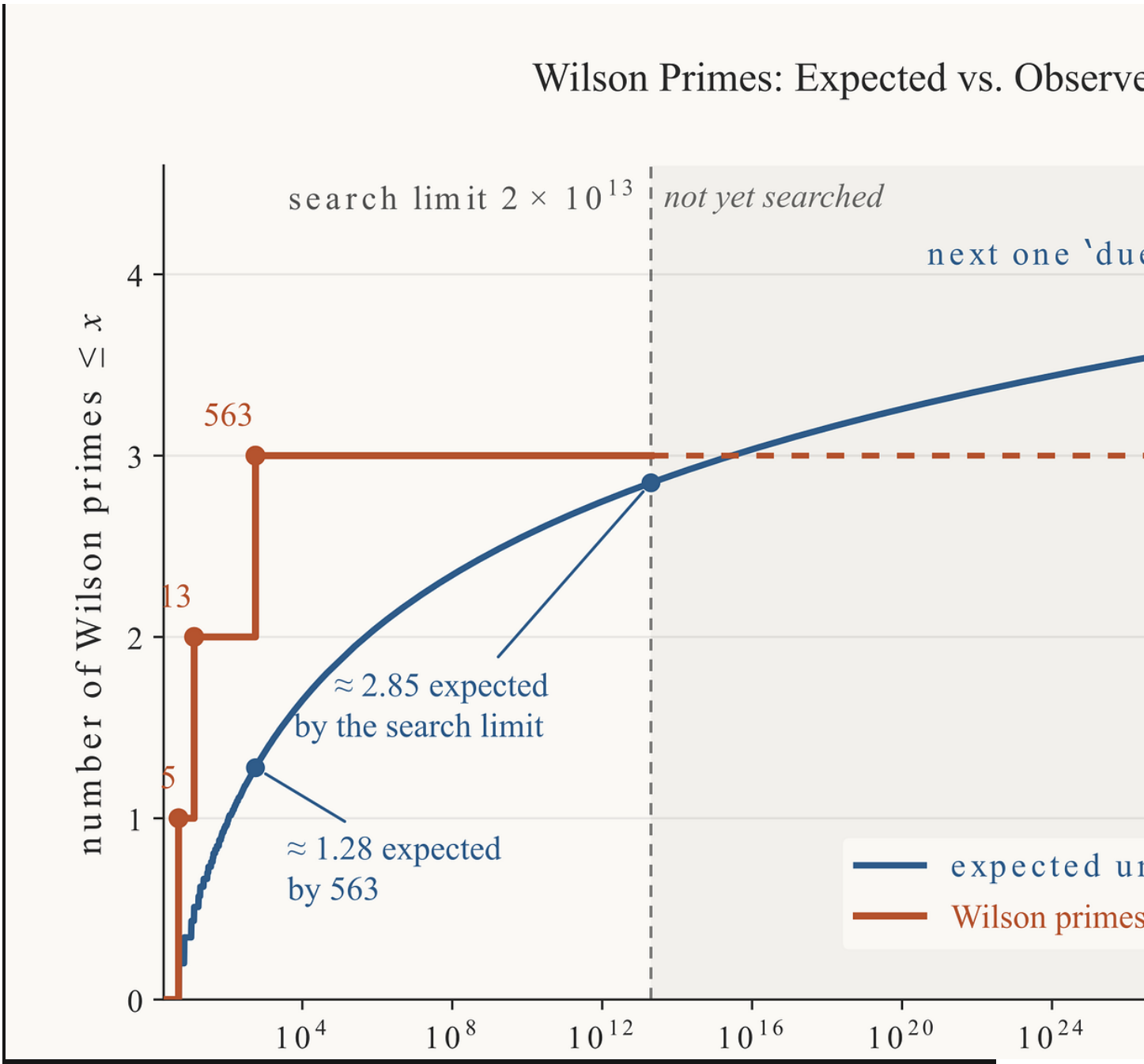
Starting from $\mathbf{2 \times 10^{13}}$, setting the expected number of additional Wilson primes equal to one gives

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$$y = (2 \times 10^{13})^e \approx 1.4 \times 10^{36}$$

That number should not be interpreted as a prediction that the next Wilson prime lies near 10^{36}. The random-residue assumption is far too crude for that.

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Observed Wilson primes compared with the expected count under the simple 1/p heuristic. The three known examples appear by p = 563, while the heuristic predicts about 2.85 by the verified search limit of 2×10^{13}. Beyond that limit, the curve is an extrapolation rather than a searched region.

Its value is conceptual. It shows how an infinite family of numbers could still appear almost nonexistent from the perspective of computation.

How Do We Search Numbers This Large?

There is another difficulty hiding in the definition. To test a Wilson prime, we apparently need

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$$(p-1)!$$

Factorials become enormous extremely quickly. For example,

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$$20! = 2\,432\,902\,008\,176\,640\,000$$

while **100!** already contains **158 decimal digits**.

For a prime in the billions or trillions, explicitly constructing the full factorial would make little sense. **Fortunately, we do not need it.**

The **Wilson-prime condition** depends only on

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$$(p-1)! \pmod{p^2}$$

So a basic modular computation can multiply the factors one at a time. If *r*_old is the current remainder and *k* is the next factor, with $k = 1, 2, \ldots, p-1$, then
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$$r_{\text{new}} = (r_{\text{old}} \cdot k) \bmod p^2$$

After each multiplication, we keep only the remainder modulo p^2.
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k	$k!$	$k!$ mod 169
1	1	1
2	2	2
3	6	6
4	24	24
5	120	120
6	720	44
7	5,040	139
8	40,320	98
9	362,880	37
10	3,628,800	32
11	39,916,800	14
12	**479,001,600**	**168**

$168 = 169 - 1$, so $12! \equiv -1 \pmod{13^2}$: 13 is a Wilson prime

Computing 12! modulo $13^2 = 169$ one factor at a time. The middle column is the factorial itself; the right column is all the computer needs to store.

This small example shows the basic idea. The factorial itself grows rapidly, but the stored remainder never exceeds $p^2 - 1$. For a prime near a trillion, that means working with numbers of about 24 digits instead of a factorial with trillions of digits.

For a search extending into the trillions, even that simple approach is not enough. **Costa**, **Gerbicz**, and **Harvey** used much more sophisticated product- and remainder-tree techniques to process large collections of primes efficiently. Their work pushed the verified search range to 2×10^{13}.

The absence of another Wilson prime is therefore not simply the result of mathematicians failing to look very far.

They have looked remarkably far.

One Extra Factor

Wilson's theorem gives us a reason that
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$$(p-1)!+1$$

must contain one factor of p whenever p is prime.

The inverse-pair argument explains it completely.

But for **5**, **13**, and **563**, that factor appears twice:
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$$p^2 \mid ((p-1)!+1)$$

No comparable argument tells us when this should happen again.

Perhaps infinitely many Wilson primes exist, scattered so thinly through the integers that enormous searches repeatedly find nothing. Or perhaps the three known examples are all there are.

At present, mathematics cannot distinguish between those possibilities.

Sometimes the difference between a theorem we understand and a mystery we do not is only one extra factor.

Cover image: A glowing path of prime numbers leads through rocky mountains to 5, 13, and 563, before continuing into a vast desert beyond. AI-generated illustration.

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